

Detecting Single-Hand Riding with Integrated Accelerometer and Gyroscope of Smartphone

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ABSTRACT

Single-hand cycling poses a safety threat with the decrement of riders' response capacity. Recognizing risky behavior by prevalently used smartphones could lead to enhanced riding safety. In this work, we propose a single-hand cycling recognition method based on motion data acquired from the three-axis accelerometer and gyroscope integrated into a handlebar-installed smartphone. We conducted a 4-person experiment. The data result demonstrates that motion data of double-hand cycling clearly distinguishes from that of single-hand, revealing the chance to materialize a robust detection tool in smartphones to enable safer biking. For future work, we prepare to redesign the experiment under more sophisticated circumstances with an improved platform, thus scaling this sensing method for real-life usage.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing.

KEYWORDS

Human Activity Recognition, Accelerometer, Gyroscope

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1 INTRODUCTION

The safety of riding trips needs to be paid added attention. Globally, 86% of bicycle travels take place on roads with extremely restricted conditions such as poor surface, lack of safe crossings, and mixed usage with traffic of velocity limitation above 60km/h¹. Due to violation of handlebar operation stipulation, more than 1544 bicycle-related accidents happened on average annually in Japan alone over

the past decade². Hence it is essential to recognize jeopardizing single-hand riding to improve the safety concern of bicyclists and other road users. It would be straightforward to place touch sensors in the handlebar to acknowledge the holding situation and even the force applied to it, but the high installation cost often makes it unpractical for individual users and uneconomical for bike-sharing providers. The prevalence of smartphones installed in the handlebar for navigation enables versatile data collecting, as well as security alert delivering. Previous work has investigated sensing methods for mobile safety [3] [2]. However, these schemes do not consider handlebar gripping.

Therefore, to fill this gap, we propose a single-hand riding detection method. Our work aims to predict the handlebar gripping scenario from the acceleration and gyro sensor integrated into the smartphone. A data-collecting platform was installed in the test bike and the collected riding behavior data help trained the machine learning model. Evaluation demonstrates the high validity of our model.

2 RELATED WORK

This work follows the category of **Cycling Behavior Recognition Schemes with Smartphones**. The rise of smartphones facilitates an increasing number of researches aimed at smartphone-based human activity behavior sensing. Wu *et al.* [1] developed BikeMate, a multi-function system that is able to study individual-level characteristics of dangerous riding behavior by a smartphone in the pocket, including retrograde riding and lane weaving. Usami *et al.* [4] utilize the embedded motion sensor of a handlebar-installed smartphone to build a recognition method of basic bicycle behavior, through noise cancellation and random forest classifier.

This paper is inspired by these previous works and complementary to them in detecting extra jeopardizing behavior of single-hand cycling with the smartphone. We use the handlebar-installed cell-phone and plan to develop a detect-in-pocket version in the future.

3 SYSTEM DESIGN

This section briefs the steps to estimate the rider's handlebar gripping scenario.

3.1 Collection Platform

Our test bike is as shown in Fig.1(A). We employed a Raspberry-Pi-based capacitive touch sensor to monitor the ground truth of the gripping situation of conductive-paint-covered handlebars. A Pixel 3a smartphone is utilized to measure the acceleration and gyro data. The installation of the smartphone is marginally inclined for better

¹WHO, <https://www.who.int/publications/i/item/9789241565684/>

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²National Police Agency, <https://www.npa.go.jp/news/release/2021/20210218jiko.html>

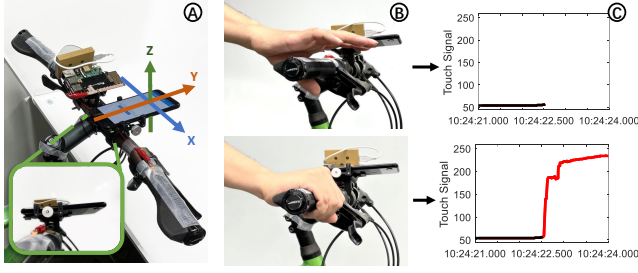


Figure 1: Data collection platform. (A) Oblique and side view of devices for riding experiment. (B) Participant releasing and gripping the handlebar. (C) Corresponding touch signal

simulation of free riding. The touch status could be recognized by the capacitance readings from the touch sensor, as illustrated in Fig.2(B) and Fig.2(C). The touch data was collected by Raspberry Pi, transmitted to a Pixel 3a smartphone via its mobile hotspot, and stored in CSV files along with motion readings. The sample rate of all sensors is 80 Hertz.

3.2 Data Processing

Intuitively, riding with only one hand would undermine the rider's control capacity over the bicycle, resulting in increment sway. Fig.2 demonstrates the feature of motion readings along and around the Z-axis of one partaker's data. Notably, for the acceleration data of single-hand riding, the power is denser over the frequency range between 10 and 28 Hertz. For rotation records, the trend is similar under 3 Hertz but subtler for else frequency range under 40 Hertz, reflecting that the single-hand-riding-related stability reduction is composed of some increment rotations and faster minor fluctuations, to which the accelerometer is more keen. Comparable patterns could be found in data along or around other axes, probably due to the inclined position of the smartphone and the intrinsic feature of riding stagger. We also notice that the zero-frequency component of acceleration data is remarkably high but contains no meaningful information. It was eliminated before further feature extraction.

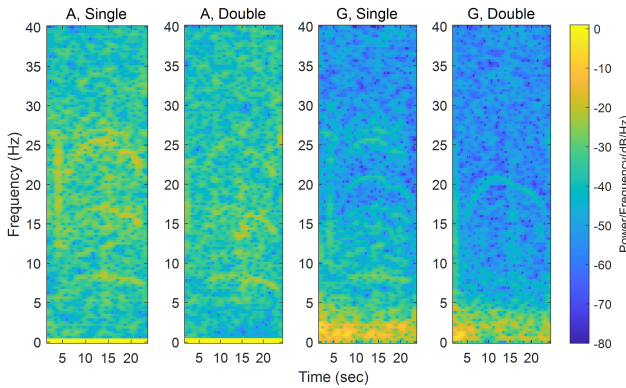


Figure 2: Spectrogram of Z-axis acceleration(A) and gyro(G) signal for double-hand and single-hand cycling

Table 1: Assessment of classification result (macro average)

Attendant	Precision	Recall	F1-score
A	0.65	0.63	0.62
B	0.85	0.85	0.85
C	0.80	0.81	0.80
D	0.91	0.89	0.89

Accordingly, we adopted a set of five extreme points with the highest amplitudes in the frequency domain to extract the feature of the 6-axis motion sensor data. The sliding window method was applied and our data was sliced into 1562 segments. Each contains 40 values, equally 0.5 seconds. Then we employed the gradient boosting machine(GBM) to distinguish the hand grasping cases.

4 EVALUATION AND PRELIMINARY RESULT

There were 4 volunteers for our experiment in total. Each was asked to ride straight forward and switch handlebar grasping several times. Attendant A rode on asphalt road and the other three on smooth bicycle-way. A range of about 40 meters long is covered for every path.

To evaluate the robustness of our model, we utilized the leave-one-out cross-validation: taking turns to test the data of each participant with the model trained by that of the other 3 persons'. The result is presented in Table 1. The average F1-score is 79%. The worst score of 62% is reached when the data of the asphalt rider is tested. This indicates that our model could efficiently classify the hand-gripping situation without noise such as different road surfaces. But it needs to gain more adaptability.

5 CONCLUSION AND FUTURE WORK

In this study, we put forward a new method to detect whether the bicycle rider is riding single-hand. We collected the 6-axis motion records from a handlebar-installed smartphone and labeled the data with ground truth from the touch sensor. The estimation model was built upon the gradient boost model and it achieves promising accuracy for experiment data. For future work, our plan includes generalizing the model under the free-riding environment and enabling the adaptability for distinguished individual riding hobbies.

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